What is Node.js

Node.js Overview

- Node.js is an open-source server environment.
- It allows you to run JavaScript on the server.

Example: Get Your Own Node.js Server

```
var http = require('http');

http.createServer(function (req, res) {
  res.writeHead(200, {'Content-Type': 'text/plain'});
  res.end('Hello World!');
}).listen(8080);
```

Node.js Reference

- Node.js has a set of built-in modules.

Download Node.js

- Download Node.js from the official Node.js website: https://nodejs.org

Node.js Introduction

- What is Node.js?
  - Node.js is an open-source server environment.
  - It is free.
  - Node.js runs on various platforms (Windows, Linux, Unix, Mac OS X, etc.).
  - It uses JavaScript on the server.

Why Node.js?

- Node.js employs asynchronous programming!
  - A typical task for a web server might be to open a file and return its content to the client.
  - Here's how PHP or ASP handles a file request:
    1. Sends the task to the computer's file system.
    2. Waits while the file system opens and reads the file.
    3. Returns the content to the client.
    4. Ready to handle the next request.
  - Here's how Node.js handles a file request:
    1. Sends the task to the computer's file system.
    2. Ready to handle the next request.
    3. Once the file system has opened and read the file, the server returns the content to the client.

    - Node.js eliminates waiting and continues with the next request.
- Node.js runs single-threaded, non-blocking, asynchronous programming, which is very memory efficient.

What Can Node.js Do?

- Generate dynamic page content.
- Create, open, read, write, delete, and close files on the server.
- Collect form data.
- Add, delete, modify data in your database.

What is a Node.js File?

- Node.js files contain tasks that will be executed on certain events.
- A typical event is someone trying to access a port on the server.
- Node.js files must be initiated on the server before having any effect.
- Node.js files have the extension ".js".

Node.js: Getting Started

- **Downloading and Installing Node.js**
    - Download Node.js from the official website: https://nodejs.org
    - Once you have downloaded and installed Node.js on your computer, let's display "Hello World" in a web browser.

Create a Node.js File

- Name it "myfirst.js" and add the following code:

```
var http = require('http');

http.createServer(function (req, res) {
  res.writeHead(200, {'Content-Type': 'text/html'});
  res.end('Hello World!');
}).listen(8080);
```

- Save the file on your computer, for example at `C:\Users\Your Name\myfirst.js`.
- The code instructs the computer to respond with "Hello World!" if anyone tries to access your computer on port 8080.

Command Line Interface

- Node.js files must be initiated through the "Command Line Interface" program on your computer.
    - For Windows users, press the *Start* or *Windows* button and search for "Command Prompt", or simply type "cmd" in the search field.

- Navigate to the folder containing "myfirst.js". The command line interface window should look like this:

```
C:\Users\Your Name>_
```

Initiate the Node.js File

- The file you created must be initiated by Node.js before any action can take place.
- Start your command line interface, type `node myfirst.js`, and press Enter.

```
C:\Users\Your Name>node myfirst.js
```

Result

- Now, your computer works as a server!
- If anyone tries to access your computer on port 8080, they will receive a "Hello World!" message.
- To see this in action, start your internet browser and navigate to: http://localhost:8080